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United States Patent	12418742
Kind Code	B2
Date of Patent	September 16, 2025
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Wireless earphone with radar capabilities

Abstract

A wireless earphone with radar capabilities includes a first earphone having a first detection unit and a container. When the first earphone is placed inside the container, a first detection wave generated by the first detection unit passes through the container. The first detection wave is reflected from a human body as a first reflected wave, and when the first detection unit receives the first reflected wave through the container, the first detection unit generates a first detection result data and sends the first detection result data to an external device. The container containing the first earphone allows a user to hold the container with the first earphone within, and since the first earphone continues to detect movements outside of the container, the container is transformed into a multifunctional controller or detector for adapting to new application modes initiated by the external device.

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Appl. No.: 18/343332

Filed: June 28, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240365042 A1	Oct. 31, 2024

Foreign Application Priority Data

TW	112115632	Apr. 26, 2023
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Publication Classification

Int. Cl.: H04R1/10 (20060101); G01S13/06 (20060101); G06F1/16 (20060101); G06F3/01 (20060101)

U.S. Cl.:

CPC H04R1/1041 (20130101); G01S13/06 (20130101); G06F1/163 (20130101); G06F3/017 (20130101); H04R2460/03 (20130101); H04R2460/07 (20130101)

Field of Classification Search

CPC: H04R (1/1041); G01S (13/06); G06F (1/163); G06F (3/017)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

(1) This application claims the priority benefit of TW application serial No. 112115632 filed on Apr. 26, 2023, the entirety of which is hereby incorporated by reference herein and made a part of the specification.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(2) The present invention relates to a wireless earphone, more particularly a wireless earphone with radar capabilities.

2. Description of the Related Art

(3) In this day and age, most commercially available wireless headphones only have a singular function of playing music or playing audio. When a user finishes using the wireless headphones and stops listening to music, the user naturally puts the wireless earphones into a container for storing and/or charging the wireless earphones. As such, the wireless earphones will automatically

switch off after a set amount of time, or rather, the wireless earphones will automatically switch off once starting charging. However, the aforementioned course of actions for using the wireless headphones are too simplistic and limiting. The wireless headphones are being used without having their full potential explored.

SUMMARY OF THE INVENTION

(4) To overcome the aforementioned problem, the present invention provides a wireless earphone with radar capabilities. The wireless earphone with radar capabilities of the present invention is able to detect movements of a user through a radar, and furthermore, the wireless earphone with radar capabilities continues to detect movements outside of a container by using the radar to generate radar waves even after being stored inside the container. As such, a new function of the wireless headphone is created after storing the wireless headphone in the container, and thus a novel application is explored for the wireless headphone.

(5) The wireless earphone with radar capabilities includes: a first earphone, having: a first communications unit, configured to wirelessly connect to an external device; a first audio unit, electrically connected to the first communications unit; wherein when the first audio unit receives an audio signal from the external device through the first communications unit, the first audio unit generates a first audio according to the audio signal; a first detection unit, electrically connected to the first communications unit, and generating a first detection wave; wherein the first detection wave is reflected from an object as a first reflected wave; when the first detection unit receives the first reflected wave, the first detection unit generates a first detection result data, and the first detection unit sends the first detection result data to the external device through the first communications unit; wherein the object is moving or vibrating as a living organism or an inorganic object; a container, configured to contain the first earphone; wherein when the first earphone is placed inside the container, the first detection unit continues to send the first detection wave and to receive the first reflected wave, and the first communications unit maintains a wireless connection to the external device.

(6) In other words, when the first earphone is placed inside the container, the first earphone of the present invention continues to generate the first detection wave that passes through the container, and continues to receive the first reflected wave that also passes through the container for continuously detecting movements outside of the container. With the first earphone inside, the container enables the user of the present invention to easily hold onto both the container and the first earphone inside of the container, while simultaneously, the first detection unit of the first earphone still continuously detects movements outside of the container from within the container. As a result, the container with the first earphone therein is transformed into a multifunctional controller or detector. This allows the first earphone and the container of the present invention to satisfy other novel functions relating to control or detection apart from just having a simple function of playing audios.

(7) In an embodiment of the present invention, after the first earphone is placed inside the container, the container containing the first earphone is transformed into a game controller. The container containing the first earphone is configured to receive controls from a user when the user plays a game displayed on the external device. In this situation, the first detection result data sent by the container containing the first earphone to the external device is a game control signal generated by the game controller.

(8) In another embodiment of the present invention, after the first earphone is placed inside the container, the container containing the first earphone is transformed into a presentation controller. The container containing the first earphone is configured to receive controls from the user when the user controls the presentation displayed on the external device. In this situation, the first detection result data sent by the container containing the first earphone to the external device is a presentation control signal generated by the presentation controller.

(9) In yet another embodiment of the present invention, after the first earphone is placed inside the

container, the container containing the first earphone is transformed into a detector for monitoring a health status of the user. The container containing the first earphone is configured to detect and monitor a breathing frequency or a pulse of the user. In this situation, the first detection result data sent by the container containing the first earphone to the external device is a health information generated by the detector for monitoring the health status of the user.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view of a wireless earphone with radar capabilities of the present invention.
- (2) FIG. 2 is a block diagram of the wireless earphone with radar capabilities of the present invention.
- (3) FIG. 3 is a flow chart for the wireless earphone with radar capabilities in an embodiment of the present invention.
- (4) FIG. 4 is a flow chart for the wireless earphone with radar capabilities in another embodiment of the present invention.
- (5) FIG. 5 is a flow chart for the wireless earphone with radar capabilities in yet another embodiment of the present invention.
- (6) FIG. 6 is a perspective view of an application of the wireless earphone with radar capabilities in an embodiment of the present invention.
- (7) FIG. 7 is a perspective view of an application of the wireless earphone with radar capabilities in another embodiment of the present invention.
- (8) FIG. 8 is a perspective view of an application of the wireless earphone with radar capabilities in yet another embodiment of the present invention.
- (9) FIG. 9 is a block diagram of a first earphone connecting to a container for the wireless earphone with radar capabilities of the present invention.
- (10) FIG. 10 is a perspective view of an application of the wireless earphone with radar capabilities in another embodiment of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

- (11) With reference to FIGS. 1 and 2, the present invention provides a wireless earphone with radar capabilities. The wireless earphone with radar capabilities includes a first earphone **10** and a container **20**. The first earphone **10** is configured to be used together with an external device **100**.
- (12) The external device **100** is a conventional device that is able to connect to wireless earphones. For example, the external device **100** is a smart phone, a computer, a tablet computer, or any other type of smart device that is able to connect to a wireless network. The external device **100** communicates with the first earphone **10** of the present invention through a computer program or an application (APP).
- (13) The first earphone **10** of the present invention has a first communications unit **11**, a first audio unit **12**, and a first detection unit **13**. The first communications unit **11** is respectively electrically connected to the first audio unit **12** and the first detection unit **13**.
- (14) The first communications unit **11** is configured to wirelessly connect to the external device **100**. In an embodiment of the present invention, the first communications unit **11** uses Bluetooth to wirelessly connect to the external device **100**. In other words, the external device **100** and the first communications unit **11** wirelessly communicate with a Bluetooth signal **40**. The external device **100** may be a conventional device that connects to wireless headphones, and may be able to generate an audio signal for the wireless headphones to play. The external device **100** sends the audio signal to the first communications unit **11** of the present invention. When the first audio unit **12** receives the audio signal from the external device **100** through the first communications unit **11**,

the first audio unit **12** generates a first audio according to the audio signal and thus the first audio unit **12** plays the first audio.

(15) The first detection unit **13** of the present invention generates a first detection wave **131**. The first detection wave **131** contacts an object and is reflected from the object as a first reflected wave **132**. When the first detection unit **13** receives the first reflected wave **132**, the first detection unit **13** generates a first detection result data, and the first detection unit **13** sends the first detection result data to the external device **100** through the first communications unit **11**. The object detected by the present invention is free to be of any kind; in other words, the object is a living organism or an inorganic object that is moving or vibrating, allowing the object to be radar-detectable. For example, in the present invention, the object detected by the present invention is a human body **200**. In another embodiment, the object may also be the living organism of any other kind, such as an animal like a cat, a dog, or a rabbit, etc. The object may also be the inorganic object, such as a toy, a motor, or a machine, etc. Of course, the object may also be a combination of the two, for example, a boy holding a baseball bat. The following embodiments use the human body **200** as the object being detected by the present invention as demonstrations of applications enabled by the present invention.

(16) The container **20** is configured to store the first earphone **10**. When the first earphone **10** is placed inside the container **20**, the first detection unit **13** continues to send the first detection wave **131** and to receive the first reflected wave **132**, and the first communications unit **11** maintains a wireless connection to the external device **100**. The container **20** of the present invention has a non-metal shell, for example, a plastic shell. As such, the first detection wave **131** generated by the first detection unit **13** is able to pass through the container **20**, and similarly, the first detection unit **13** is able to receive the first reflected wave **132** that passes through the container **20**. In the present embodiment, the first detection unit **13** includes antennas for transmitting and receiving electromagnetic waves as radar signals. The first detection wave **131** transmitted by the first detection unit **13** and the first reflected wave **132** received by the first detection unit **13** are both millimeter waves (mmWaves). After the first earphone **10** is placed inside the container **20**, the first detection wave **131** and the first reflected wave **132** as millimeter waves are generally unaffected by the container **20**. Without being blocked by the container **20**, the first detection wave **131** and the first reflected wave **132** are used by the first detection unit **13** for detecting movements outside of the container **20**.

(17) After the first earphone **10** is placed inside the container **20**, the first earphone **10** of the present invention continues to generate the first detection wave **131** that passes through the container **20**, and continues to receive the first reflected wave **132** that also passes through the container **20** for continuously detecting movements outside of the container **20**. With the first earphone **10** inside, the container **20** enables a user of the present invention to easily hold onto both the container **20** and the first earphone **10** inside of the container **20**, while simultaneously, the first detection unit **13** of the first earphone **10** still continuously detects movements outside of the container **20** from within the container **20**. As a result, the container **20** with the first earphone **10** therein is transformed into a multifunctional controller or detector. This allows the first earphone **10** and the container **20** of the present invention to satisfy other novel functions relating to control or detection apart from just having a simple function of playing audios.

(18) Furthermore, the first detection unit **13** is a unit capable of detecting body movements of the user with millimeter radar waves. A hardware of the first detection unit **13** is a system on a chip (SoC) that integrates radar antennas, a memory, and a processor. The memory integrated within the first detection unit **13** stores a pre-trained artificial intelligence (AI) detection model. The pre-trained AI detection model is able to help identifying body movements of the user detected by the radar waves, and thus help the first detection unit **13** to generate the first detection result data. For the pre-trained AI detection model stored in the memory inside the first detection unit **13**, weights used by the pre-trained AI detection model are able to be updated. In an embodiment, the weights

used by the pre-trained AI detection model are updated by the external device **100**. In other words, when the first communications unit **11** is communicatively connected to the external device **100**, the pre-trained AI detection model stored in the first detection unit **13** is updated by the external device **100**.

(19) With reference to FIG. **3**, in the current embodiment, a wall of the container **20** minimally reflects the first detection wave **131**. As such, the first reflected wave **132** received by the first detection unit **13** also reflects information of being partially reflected from the wall of the container **20**. This information will be present in the first detection result data generated by the first detection unit **13** according to the first reflected wave **132**. The first detection unit **13** of the first earphone **10** executes the following steps:

(20) Step **S300**: determining whether the first earphone **10** is placed inside the container **20** according to the first detection result data. For instance, in an embodiment, determining whether the first detection result data presents an information that a distance the first reflected wave **132** is reflected off equals a particular distance, wherein the particular distance is a given distance between the first earphone **10** and the wall of the container **20**; if yes, then determining that the first earphone **10** is placed inside the container **20**, or else, determining that the first earphone **10** is still outside of the container **20**.

(21) Step **S310**: when determining that the first earphone **10** is still outside of the container **20**, not generating a mode change signal by the first detection unit **13**.

(22) Step **S320**: when determining that the first earphone **10** is placed inside the container **20**, generating the mode change signal by the first detection unit **13**, and sending the mode change signal to the external device **100** through the communications unit **11**.

(23) Step **S330**: maintaining transmitting the first detection wave **131** and receiving the first reflected wave **132** for generating the first detection result data, and maintaining sending the first detection result data to the external device **100** that is wirelessly connected through the communications unit **11**.

(24) With reference to FIGS. **1** and **2**, in another embodiment, the present invention further includes a second earphone **30**. The first earphone **10** and the second earphone **30** form a pair of wireless earphones, as the first earphone **10** is a primary earphone and the second earphone **30** is a secondary earphone. The primary earphone may be either a left ear wireless earphone or a right ear wireless earphone, and the secondary earphone is a corresponding wireless earphone on an opposite side to the primary earphone. The first earphone **10** and the second earphone **30** are connected via Bluetooth; more particularly, the first earphone **10** is respectively connected to the second earphone **30** and the external device **100** by using True Wireless Stereo (TWS). The first communications unit **11** of the first earphone **10** sends the Bluetooth signal to the second earphone **30** for communicatively connecting to the second earphone **30**.

(25) The second earphone **30** includes a second communications unit **31**, a second audio unit **32**, and a second detection unit **33**. The second communications unit **31** is respectively electrically connected to the second audio unit **32** and the second detection unit **33**.

(26) The second communications unit **31** is wirelessly connected to the first communications unit **11** for communicating to the first earphone **10**. When the first communications unit **11** receives the audio signal from the external device **100**, the first communications unit **11** sends the audio signal to the second communications unit **31**. When the second audio unit **32** receives the audio signal from the first communications unit **11** through the second communications unit **31**, the second audio unit **32** generates a second audio according to the audio signal and thus the second audio unit **32** plays the second audio. For example, the first audio played by the first audio unit **12** and the second audio played by the second audio unit **32** are respectively a left ear soundtrack and a right ear soundtrack played by the external device **100** through the present invention.

(27) The second detection unit **33** generates a second detection wave **331**. The second detection wave **331** contacts the object, such as the human body **200**, and is reflected from the human body

200 as a second reflected wave **332**. When the second detection unit **33** receives the second reflected wave **332**, the second detection unit **33** generates a second detection result data, and the second detection unit **33** sends the second detection result data to the first communications unit **11** through the second communications unit **31**. When the first communications unit **11** receives the second detection result data from the second communications unit **31**, the first communications unit **11** sends the second detection result data to the external device **100**.

(28) The container **20** is also configured to store the second earphone **30**. When the second earphone **30** is placed inside the container **20**, the second detection unit **33** continues to send the second detection wave **331** and to receive the second reflected wave **332**, and the second communications unit **31** maintains a wireless connection to the first communications unit **11**. The second detection wave **331** generated by the second detection unit **33** is also able to pass through the container **20**, and similarly, the second detection unit **33** is able to receive the second reflected wave **332** that passes through the container **20**.

(29) As such, when both the first earphone **10** and the second earphone **30** are placed inside the container **20**, the first earphone **10** and the second earphone **30** respectively generate the first detection wave **131** and the second detection wave **331**, and the first earphone **10** and the second earphone **30** respectively receive the first reflected wave **132** and the second reflected wave **332** for respectively generating the first detection result data and the second detection result data. In an embodiment, the first earphone **10** and the second earphone **30** are fixed facing two different directions inside of the container **20**, and therefore the first earphone **10** and the second earphone **30** are generating and receiving electromagnetic waves at different directions. Since the first earphone **10** and the second earphone **30** are detecting movements in different directions, the first detection result data and the second detection result data are independent from each other. The second detection wave **331** generated from the second detection unit **33** and the second reflected wave **332** received by the second detection unit **33** are also millimeter waves.

(30) With reference to FIG. 4, in the present embodiment, the wall of the container **20** minimally reflects the second detection wave **331**. As such, the second reflected wave **332** received by the second detection unit **33** also reflects information of being partially reflected from the wall of the container **20**. This information will also be present in the second detection result data generated by the second detection unit **33** according to the second reflected wave **132**.

(31) After receiving the second detection result data, the first communications unit **11** of the first earphone **10** executes the following steps:

(32) Step **S400**: determining whether the first earphone **10** and the second earphone **30** are simultaneously present inside of the container **20** according to the first detection result data and the second detection result data. In this embodiment, the first communications unit **11** determines whether the second earphone **30** is placed inside the container **20** by using the same method that the first detection unit **13** determines whether the first earphone **10** is placed inside the container **20**.

(33) Step **S410**: when determining that either the first earphone **10** or the second earphone **30** is still outside of the container **20**, not sending out the mode change signal from the first communications unit **11**.

(34) Step **S420**: when determining the first earphone **10** and the second earphone **30** are simultaneously present inside of the container **20**, sending the mode change signal from the first communications unit **11** to the external device **100**.

(35) Step **S430**: maintaining the wireless connections to the external device **100** and the second communications unit **31** for sending the first detection result data and the second detection result data to the external device **100**.

(36) With reference to FIGS. 2 and 5, in an embodiment of the present invention, the first earphone **10** further includes a first battery unit **14**, and the second earphone **30** further includes a second battery unit **34**. The first battery unit **14** is respectively electrically connected to the first communications unit **11**, the first audio unit **12**, and the first detection unit **13**. The second battery

unit **34** is respectively electrically connected to the second communications unit **31**, the second audio unit **32**, and the second detection unit **33**. The first battery unit **14** holds a first earphone charge, and the second battery unit **34** holds a second earphone charge.

(37) The container **20** includes a recharge socket and a recharge battery. The recharge battery holds a recharge station charge, and the recharge battery is electrically connected to the recharge socket. The recharge socket is configured to respectively electrically connect to the first battery unit **14** of the first earphone **10** and the second battery unit **34** of the second earphone **30**.

(38) When the recharge socket is electrically connected to the first battery unit **14**, the recharge battery starts recharging the first battery unit **14** with the recharge station charge through the recharge socket, and the first battery unit **14** generates a first recharging signal to the first communications unit **11**. Similarly, when the recharge socket is electrically connected to the second battery unit **34**, the recharge battery starts recharging the second battery unit **34** with the recharge station charge through the recharge socket, and the second battery unit **34** generates a second recharging signal to the second communications unit **31**. When the second communications unit **31** receives the second recharging signal from the second battery unit **34**, the second communications unit **31** sends the second recharging signal to the first communications unit **11**.

(39) The first communications unit **11** of the first earphone **10** executes the following steps:

(40) Step **S500**: determining whether receiving the first recharging signal from the first battery unit **14**.

(41) Step **S510**: when without receiving the first recharging signal from the first battery unit **14**, not generating the mode change signal by the first communications unit **11**.

(42) Step **S520**: when receiving the first recharging signal from the first battery unit **14**, further determining whether the first detection result data is a power off command.

(43) Step **S530**: when determining the first detection result data is the power off command, powering off the first earphone **10** and the second earphone **30**.

(44) Step **S540**: when determining the first detection result data is otherwise than the power off command, further determining whether simultaneously receiving the second recharging signal from the second communications unit **31**; when determining without simultaneously receiving the second recharging signal from the second communications unit **31**, executing step **S510**.

(45) Step **S550**: when determining simultaneously receiving the first recharging signal from the first battery unit **14** and receiving the second recharging signal from the second communications unit **31**, determining that the first earphone **10** and the second earphone **30** are simultaneously placed inside of the container **20** and thus generating the mode change signal and sending the mode change signal to the external device **100**.

(46) Step **S560**: maintaining the wireless connections to the external device **100** and the second communications unit **31** for sending the first detection result data and the second detection result data to the external device **100**.

(47) In the current embodiment, after the user places the first earphone **10** and the second earphone **30** inside of the container **20**, the user is able to make a hand gesture to power off the first earphone **10** and the second earphone **30**. However, if the user lets the first earphone **10** and the second earphone **30** stay powered, the first earphone **10** will generate the mode change signal and send the mode change signal to the external device **100**. As a result, the first earphone **10** and the second earphone **30** will continue to function under a new application mode. This way, the first earphone **10** and the second earphone **30**, under the new application mode set by the external device **100**, will bring forth the user of the present invention new experiences unrelated to just simply playing audios.

(48) With reference to FIG. **6**, in the current embodiment, the first detection unit **13** and the second detection unit **33** are respectively facing a left side and a right side outside of the container **20**. The first detection unit **13** and the second detection unit **33** respectively generate the first detection wave **131** and the second detection wave **331** aimed at hands of the human body **200** of the user.

This way, the first detection unit **13** and the second detection unit **33** are able to respectively detect the first reflected wave **132** and the second reflected wave **332** respectively reflected from a left hand gesture **210** and a right hand gesture **220** posed by the user. As a result, the first detection unit **13** generates the first detection result data according to the first reflected wave **132** corresponding to the left hand gesture **210**, and the second detection unit **33** generates the second detection result data according to the second reflected wave **332** corresponding to the right hand gesture **220**. The first detection result data and the second detection result data are respectively a hand gesture detection result data.

(49) When the external device **100** in cooperation with the present invention receives the mode change signal from the first communications unit **11**, the external device **100** changes an application mode for cooperating with the present invention. The change of application mode can be, for example, switching mode from using the first earphone **10** and the second earphone **30** of the present invention to play music or play audio to other application modes such as using the first earphone **10** and the second earphone **30** of the present invention to play games, to control presentation slides, and to monitor health, etc. The present invention is multifunctional in its ability to cooperate with the external device **100** for various applications. Application modes initiated by the external device **100** on the present invention is free to be otherwise than the few examples mentioned in the present invention.

(50) In the present embodiment, after the first earphone **10** and the second earphone **30** are placed in the container **20**, the container **20** with the first earphone **10** and the second earphone **30** placed within is transformed into a game controller. The said game controller is configured to detect the left hand gesture **210** and the right hand gesture **220** posed by the user on opposite sides outside of the container **20** with radar waves generated by the first earphone **10** and the second earphone **30** inside of the container **20**. Such detection of the left hand gesture **210** and the right hand gesture **220** allows the user to move hands for controlling the game displayed on the external device **100**. In this situation, the first detection result data and the second detection result data sent by the present invention to the external device **100** are game control signals generated by the game controller. In the example shown in FIG. **6**, the container **20** with the first earphone **10** and the second earphone **30** placed within is transformed into a virtual steering wheel for controlling a car racing game, and the user is able to rotate a direction of the virtual steering wheel by moving hand gestures.

(51) With reference to FIG. **7**, in another embodiment of the present invention, after the first earphone **10** is placed inside the container **20**, the container **20** with the first earphone **10** within is transformed into a presentation controller. The container **20** as the presentation controller is configured to receive controls from the user as the user controls the presentation displayed on the external device **100**. In this situation, the first detection result data sent by the present invention to the external device **100** is a presentation control signal generated by the presentation controller. For example, the first detection unit **13** of the first earphone **10** sends out the first detection wave **131** towards a right hand of the user for detecting the right hand gesture **220**. The user moves the right hand gesture **220** upwards next to the container **20** with the first earphone **10** within for switching to a previous slide of the presentation, and the user moves the right hand gesture **220** downwards next to the container **20** for switching to a next slide of the presentation.

(52) With reference to FIG. **8**, in yet another embodiment of the present invention, the object being detected is the living organism, and the living organism has a torso of an animal. The first detection unit **13** sends out the first detection wave **131** aimed towards the torso of the living organism, and thus the first detection unit **13** receives the first reflected wave **132** reflected from the torso of the living organism. More particularly, for example, the first detection unit **13** sends out the first detection wave **131** aimed towards a chest **230** or a back of the human body **200**, and thus the first detection unit **13** receives the first reflected wave **132** reflected from the chest **230** or the back of the human body **200**. As such, the first detection result data generated by the first detection unit **13**

is a measurement of movements of the torso of the living organism, and this measurement entails a vital signs data of the living organism. Of course, the present invention may also aim at other body parts of the living organism for detecting the vital signs data of the living organism. The vital signs data of the living organism, for example, includes a breathing frequency data or a pulse data. After the first earphone **10** is placed inside the container **20**, the container **20** containing the first earphone **10** is transformed into a detector for monitoring a health status of the user. The container **20** as the detector is configured to detect and monitor a breathing frequency or a pulse of the user. In this situation, the first detection result data sent by the present invention to the external device **100** is a health information generated by the detector for monitoring the health status of the user. For example, the user breathes normally beside the container **20** containing the first earphone **10**, and the first earphone **10** within the container **20** continuously transmits the first detection wave **131** to detect rise and fall movements or jumping movements of the chest **230** of the user outside of the container **20**. The first earphone **10** collects detected data to generate the health information to send to the external device **100**, and the external device **100** keeps records and logs of the health information for the user.

(53) With further reference to FIG. **9**, in the current embodiment, the container **20** further includes a container switch **21** and a container communications unit **22**. The container communications unit **22** is wirelessly connected to the first communications unit **11** of the first earphone **10** by using Bluetooth, and the container communications unit **22** is electrically connected to the container switch **21**. The container switch **21** is mounted on the container **20**, thus allowing the user to switch on or switch off the container switch **21**. The container switch **21** is free to take any form as a switch. In the current embodiment, the container switch **21** is a button. In other embodiments, the container switch **21** may be a tumbler switch.

(54) When the container switch **21** is switched on, the container switch **21** generates a container signal to the container communications unit **22**, and the container communications unit **22** sends the container signal to the first communications unit **11** of the first earphone **10**. The first communications unit **11** determines whether to send the mode change signal to the external device **100**, in accordance to a mode changing mechanism configured for the present invention, according to whether the container signal is received by the first communications unit **11** from the container communications unit **22**.

(55) For example, after the first communications unit **11** generates the mode change signal, the first communications unit **11** further determines whether receiving the container signal from the container communications unit **22** of the container **20**.

(56) When the first communications unit **11** determines without receiving the container signal, the first communications unit **11** refrains from sending the mode change signal to the external device **100**. When the first communications unit **11** determines receiving the container signal, the first communications unit **11** proceeds to send the mode change signal to the external device **100**. As such, by switching on or switching off the container switch **21** on the container **20**, the user of the present invention can decide whether to enable the aforementioned mode change function of having the container **20**, after containing the first earphone **10** and the second earphone **30**, sending out the mode change signal to the external device **100** for changing the application mode of using the present invention.

(57) Furthermore, with reference to FIG. **10**, in another embodiment of the present invention, after the first earphone **10** is placed inside the container **20**, the container **20** with the first earphone **10** within is transformed into a controller of a personal device. The said personal device is connected to the first earphone **10** of the present invention through TWS Bluetooth, and the said personal device can be any device with Bluetooth communication capabilities, such as a smart phone **101**, a smart watch **102**, a tablet computer **103**, or a pair of smart glasses **104**. The container **20** with the first earphone **10** is configured to receive controls from the user as the user controls the external device **100** that is one of the aforementioned personal devices. In this situation, the first detection

result data sent by the present invention to the external device **100** is a command signal remotely controlling the aforementioned personal device.

(58) In an example, the first detection unit **13** of the first earphone **10** sends out the first detection wave **131** towards a right hand of the user for detecting the right hand gesture **220**. For example, the user may move the right hand and make the right hand gesture **220** point at a direction for remotely using a keyboard on the said personal device. Alternatively, the user may also simply move the right hand for remotely using the keyboard on the said personal device. The keyboard on the said personal device is free to have any form, for example, the said keyboard can be a keyboard of a game controller, a keyboard of a computer, a button on a touchpad, or a keyboard of any kind of device with Bluetooth communication capabilities. A movement of the right hand from the user corresponds to an application using the keyboard. For example, the movement of waving the right hand to a right side from the user corresponds to an action of pressing a right arrow button on the game controller, and the movement of waving the right hand to a left side from the user corresponds to another action of pressing a left arrow button on the game controller.

(59) With reference to FIG. 7, in another example, the present invention may also replace a function of a user being beside a pointing device and using the pointing device to control a cursor **111**. For example, the present invention provides a function to replace a function of the touchpad **110**. When the user moves the right hand, the movement of the right hand corresponds to an action of the user remotely moving a location of the cursor **111**. When the user uses the right hand to select or to click, the movement of the right hand corresponds to an action of the user remotely selecting or clicking the cursor **111**. In other words, the user is free to be away from the touchpad **110** and still be able to fully and remotely use all functions of the touchpad **110**. The said touchpad **110** may be a standalone device as the external device **100** with Bluetooth communication capabilities. With reference to FIG. 7, the said touchpad **110** may also be embedded as a part of the said personal device which is the external device **100**.

(60) In the case that the external device **100** is the smart phone **101**, in an embodiment, the smart phone **101** can also be transformed by the present invention into a command relay for the container **20** with the first earphone **10**. In this case, multiple networks of personal devices are connected to the smart phone **101** or the tablet computer **103**, for instance, the smart watch **102** is communicatively connected to the smart phone **101** or the tablet computer **103** through Bluetooth, and the pair of smart glasses **104** and a smart television **105** are respectively communicatively connected to the smart phone **101** or the tablet computer **103** through a Wi-Fi wireless network. The container **20** with the first earphone **10** generates the command signal aimed for the smart watch **102** to the smart phone **101** or the tablet computer **103**, and the smart phone **101** or the tablet computer **103** then relays the command signal to the smart watch **102**. The container **20** with the first earphone **10** may also generate the command signal as a display request to the smart phone **101** or the tablet computer **103**, and the smart phone **101** or the tablet computer **103** then relays the display request to the pair of smart sunglasses **104** or the smart television **105**, requesting the pair of smart sunglasses **104** or the smart television **105** to display a current screen on the smart phone **101** or the tablet computer **103** with Miracast over the Wi-Fi wireless network. As such, the present invention opens many possibilities for personal devices of the user to be controlled involving the participation of the container **20** with the first earphone **10**. Applications enabled by the present invention are free to involve controls of the personal devices more than just controlling keyboard actions, cursor actions and requesting display with Miracast.

(61) Furthermore, in an embodiment, the mode change signal is generated by the external device **100**. As such, the user may use the external device **100** for allowing the external device **100** to initiate a mode change, and further allow the first earphone **10** and the container **20** to be differently used for different applications. In other words, the mode change signal is generated by the external device **100**. The first detection unit **13** receives the mode change signal from the external device **100** through the first communications unit **11**, and the first detection unit **13**

changes a meaning of the first detection result data according to the mode change signal received. The external device, capable of generating the mode change signal, is the aforementioned smart device such as the smart phone **101**, the smart watch **102**, the tablet computer **103**, or the pair of smart glasses **104**. For example, the user first stores the first earphone **10** inside of the container **20**, then the user uses the external device **100** allowing the external device **100** to generate the mode change signal. When the external device **100** generates the mode change signal, for example, the first earphone **10** and the container **20** switch from being used as the aforementioned game controller to the aforementioned presentation controller. This way a mode of how the first earphone **10** and the container **20** are applied to the external device **100** is changed by the external device **100**. As such, the meaning of the first detection result data is changed according to the mode change signal. In the present example, the first detection result data is changed for representing a control signal of a game to a control signal for controlling the presentation slides.

(62) Furthermore, in another embodiment of the present invention, the smart device that works in conjuncture with the present invention is simultaneously executing multiple APPs. When either the first earphone **10** or the second earphone **30** is placed in the container **20**, the present invention generates the mode change signal, and sends the mode change signal to the external device **100** through the first earphone **10**. This way, the present invention changes an application of how the external device **100** is utilized with either the first earphone **10** or the second earphone **30**. In other words, in the present embodiment, after the first earphone **10** sends the mode change signal to the external device **100**, the first earphone **10** is utilized corresponding to one of the APPs executed by the external device **100**, and simultaneously, the second earphone **30** is utilized corresponding to another one of the APPs executed by the external device **100**. For example, the smart device is the smart phone **101** or the tablet computer **103**, and one of the APPs executed by the smart device is a game, another one of the APPs executed by the smart device is a music player. When the smart device receives the mode change signal from the first earphone **10**, the smart device utilizes either the first earphone **10** or the second earphone **30** of the present invention for continuously playing audio from the music player APP, and the smart device also utilizes the other one of the first earphone **10** or the second earphone **30**, of which is now contained in the container **20**, for switching the mode from playing audio from the game in the game APP to controlling the game as the game controller. Please be reminded that the present invention allows the first earphone **10** and the second earphone **30** to be freely utilized with the external device **100**, rather than imposing limitations to how the first earphone **10** and the second earphone **30** should be utilized with the external device **100**. The present invention also allows the external device **100** to freely execute APPs or programs as the external device **100** itself intends.

(63) The means to determine whether the first earphone **10** or the second earphone **30** is placed in the container **20** has already been stated in aforementioned paragraphs. In other words, in different embodiments of the present invention, the present invention already reveals how to determine whether the first earphone **10** and the second earphone **30** are both contained in the container **20**. The only difference being, in the present embodiment, the present invention only determines whether either the first earphone **10** or the second earphone **30** is placed in the container **20**.

(64) For example, in an embodiment, when the first communications unit **11** receives the first recharging signal from the first battery unit **14** or receives the second recharging signal from the second communications unit **31**, the first communications unit **11** determines that either the first earphone **10** or the second earphone **30** is placed in the container **20**, and thus the first communications unit **11** generates the mode change signal to send to the external device **100**.

(65) For example, in another embodiment, when the first communications unit **11** determines whether the first earphone **10** or the second earphone **30** is placed in the container **20** correspondingly according to the first detection result data or the second detection result data, regardless of whether the first earphone **10** or the second earphone **30** needs to reconnect to the external device **100** according to communications protocol configured under TWS settings, the first

earphone **10** would be the one to generate the mode change signal to the external device **100** for switching a mode of one of the earphones. This way, the first earphone **10** and the second earphone **30**, as two earphones, are able to be independently utilized with the external device **100** for running two different APPs on the external device **100**.

Claims

1. A wireless earphone with radar capabilities, comprising: a first earphone, having: a first communications unit, configured to wirelessly connect to an external device; a first audio unit, electrically connected to the first communications unit; wherein when the first audio unit receives an audio signal from the external device through the first communications unit, the first audio unit generates a first audio according to the audio signal; a first detection unit, electrically connected to the first communications unit, and generating a first detection wave; wherein the first detection wave is reflected from an object as a first reflected wave; when the first detection unit receives the first reflected wave, the first detection unit generates a first detection result data, and the first detection unit sends the first detection result data to the external device through the first communications unit; wherein the object is moving or vibrating as a living organism or an inorganic object; and a first battery unit, electrically connected to the first communications unit, the first audio unit, and the first detection unit, and holding a first earphone charge; a container, configured to contain the first earphone; wherein the container further comprises: a recharge socket, configured to electrically connect to the first battery unit of the first earphone; and a recharge battery, electrically connect to the recharge socket, and holding a recharge station charge; wherein when the recharge socket is electrically connected to the first battery unit, the recharge battery recharges the first battery unit with the recharge station charge through the recharge socket, and the first battery unit generates a first recharging signal to the first communications unit; wherein when the first communications unit receives the first recharging signal from the first battery unit, the first communications unit determines the first earphone is placed in the container, and thus the first communications unit generates a mode change signal and sends the mode change signal to the external device; and wherein when the first earphone is placed inside the container, the first detection unit continues to send the first detection wave and to receive the first reflected wave, and the first communications unit maintains a wireless connection to the external device.
2. The wireless earphone with radar capabilities as claimed in claim 1, wherein the first detection unit determines whether the first earphone is placed in the container according to the first detection result data; wherein when the first detection unit determines the first earphone is placed in the container, the first detection unit generates the mode change signal, and sends the mode change signal to the external device through the first communications unit.
3. The wireless earphone with radar capabilities as claimed in claim 1, further comprising a second earphone; wherein the second earphone further comprises: a second communications unit, wirelessly connected to the first communications unit; wherein when the first communications unit receives the audio signal from the external device, the first communications unit sends the audio signal to the second communications unit; a second audio unit, electrically connected to the second communications unit; wherein when the second audio unit receives the audio signal from the first communications unit through the second communications unit, the second audio signal generates a second audio according to the audio signal; a second detection unit, electrically connected to the second communications unit, and generating a second detection wave; wherein the second detection wave is reflected from the object as a second reflected wave; when the second detection unit receives the second reflected wave, the second detection unit generates a second detection result data, and the second detection unit sends the second detection result data to the first communications unit through the second communications unit; wherein when the first communications unit receives the second detection result data from the second communications

unit, the first communications unit sends the second detection result data to the external device; wherein the container is also configured to contain the second earphone; when the second earphone is placed inside of the container, the second detection unit continues to send the second detection wave and to receive the second reflected wave, and the second communications unit maintains the wireless connection to the first communications unit.

4. The wireless earphone with radar capabilities as claimed in claim 3, wherein the first communications unit respectively determines whether the first earphone and the second earphone are placed simultaneously in the container according to the first detection result data and the second detection result data; wherein when the first communications unit determines the first earphone and the second earphone are simultaneously placed in the container, the first communications unit generates the mode change signal, and sends the mode change signal to the external device.

5. The wireless earphone with radar capabilities as claimed in claim 3, wherein: the second earphone further comprises: a second battery unit, electrically connected to the second communications unit, the second audio unit, the recharge socket of the container, and the second detection unit, and holding a second earphone charge; when the recharge socket is electrically connected to the second battery unit, the recharge battery recharges the second battery unit with the recharge station charge through the recharge socket, and the second battery unit generates a second recharging signal to the second communications unit; when the second communications unit receives the second recharging signal from the second battery unit, the second communications unit sends the second recharging signal to the first communications unit; when the first communications unit receives the first recharging signal from the first battery unit, and also receives the second recharging signal from the second communications unit, the first communications unit determines the first earphone and the second earphone are placed in the container, and thus the first communications unit generates the mode change signal and sends the mode change signal to the external device.

6. The wireless earphone with radar capabilities as claimed in claim 5, wherein the container further comprises: a container communications unit, communicatively connected to the first communications unit; a container switch, electrically connected to the container communications unit, and mounted on the container; wherein when the container switch is switched on, the container switch generates a container signal to the container communications unit, and the container communications unit sends the container signal to the first communications unit; wherein the first communications unit determines whether to send the mode change signal to the external device according to whether the container signal is received from the container communications unit.

7. The wireless earphone with radar capabilities as claimed in claim 3, wherein the object detected by the first detection wave and the second detection wave is a human body; wherein the first detection unit and the second detection unit respectively transmit the first detection wave and the second detection wave aimed towards hands of the human body, and the first detection unit and the second detection unit respectively receive the first reflected wave and the second reflected wave reflected from a left hand gesture and a right hand gesture; wherein the first detection result data and the second detection result data are respectively a hand gesture detection result data.

8. The wireless earphone with radar capabilities as claimed in claim 3, wherein the first communications unit is wirelessly connected to the external device and the second communications unit with Bluetooth; wherein the first detection wave generated by the first detection unit, the first reflected wave received by the first detection unit, the second detection wave generated by the second detection unit, and the second reflected wave received by the second detection unit are all millimeter waves.

9. The wireless earphone with radar capabilities as claimed in claim 3, wherein: the second earphone further comprises: a second battery unit, electrically connected to the second communications unit, the second audio unit, the recharge socket of the container, and the second

detection unit, and holding a second earphone charge; when the recharge socket is electrically connected to the second battery unit, the recharge battery recharges the second battery unit with the recharge station charge through the recharge socket, and the second battery unit generates a second recharging signal to the second communications unit; when the second communications unit receives the second recharging signal from the second battery unit, the second communications unit sends the second recharging signal to the first communications unit; when the first communications unit receives the second recharging signal from the second communications unit, the first communications unit determines the second earphone is placed in the container, and thus the first communications unit generates the mode change signal and sends the mode change signal to the external device.

10. The wireless earphone with radar capabilities as claimed in claim 1, wherein the object detected by the first detection wave is a human body; wherein the first detection unit transmits the first detection wave aimed towards a hand of the human body, and the first detection unit receives the first reflected wave reflected from a left hand gesture or a right hand gesture; wherein the first detection result data is a hand gesture detection result data.

11. The wireless earphone with radar capabilities as claimed in claim 1, wherein the object detected by the first detection wave is the living organism; wherein the first detection unit transmits the first detection wave aimed towards the living organism, and the first detection unit receives the first reflected wave reflected from the living organism; wherein the first detection result data is a vital signs data of the living organism.

12. The wireless earphone with radar capabilities as claimed in claim 1, wherein the first communications unit is wirelessly connected to the external device with Bluetooth; wherein the first detection wave generated by the first detection unit and the first reflected wave received by the first detection unit are both millimeter waves.

13. The wireless earphone with radar capabilities as claimed in claim 1, wherein the first communications unit determines whether the first detection result data is a power off command; when the first communications unit determines the first detection result data is the power off command, the first communications unit powers off the first earphone.

14. The wireless earphone with radar capabilities as claimed in claim 1, wherein the container further comprises: a container communications unit, communicatively connected to the first communications unit; a container switch, electrically connected to the container communications unit, and mounted on the container; wherein when the container switch is switched on, the container switch generates a container signal to the container communications unit, and the container communications unit sends the container signal to the first communications unit; wherein the first communications unit determines whether to send the mode change signal to the external device according to whether the container signal is received from the container communications unit.

15. The wireless earphone with radar capabilities as claimed in claim 1, wherein the first communications unit is wirelessly connected to the external device with Bluetooth; wherein the first detection result data is a command signal to remote control the external device.

16. The wireless earphone with radar capabilities as claimed in claim 1, wherein the first communications unit is wirelessly connected to the external device with Bluetooth, and the external device is a smart phone or a tablet computer; wherein the first detection result data is a command signal to remote control a smart watch, a pair of smart glasses, or a smart television, and the external device that receives the first detection result data is a command relay; wherein the smart phone or the tablet computer is communicatively connected to the smart watch, the pair of smart glasses, or the smart television for correspondingly relaying the command signal to the smart watch, the pair of smart glasses, or the smart television.

17. The wireless earphone with radar capabilities as claimed in claim 1, wherein the first communications unit is wirelessly connected to the external device with Bluetooth, and the external

device is a pointing device; wherein the first detection result data is a command signal to remote control a cursor of the pointing device.

18. The wireless earphone with radar capabilities as claimed in claim 1, wherein the first communications unit is wirelessly connected to the external device with Bluetooth; wherein the first detection unit receives the mode change signal from the external device through the first communications unit, and the first detection unit changes a meaning of the first detection result data according to the mode change signal.

19. The wireless earphone with radar capabilities as claimed in claim 1, wherein the object detected by the first detection wave is the living organism; wherein the first detection unit transmits the first detection wave aimed towards the living organism, and the first detection unit receives the first reflected wave reflected from the living organism; wherein the first detection result data is a breathing frequency data or a pulse data of the living organism.
